

Foto: Dr. Matthias Otte

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**EUROfusion**

**HELMHOLTZ**

RESEARCH FOR GRAND CHALLENGES



P. Scholl<sup>1,2</sup>

# IMPURITY TRANSPORT AND RADIATION AT THE STELLARATOR WENDELSTEIN 7-X

Promotionskolloquium

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# MOTIVATION AND CONTEXT

# NUCLEAR FUSION: THE ENERGY CHALLENGE

- D-T fusion reaction:  ${}^2\text{H} + {}^3\text{H} \rightarrow {}^4\text{He}$   
(3.5 MeV) + n (14.1 MeV)
- Lawson criterion for net energy gain:

$$n_e T \tau_E \geq \frac{12 f_{\text{tot}}}{\langle \sigma_{\text{DT}} v \rangle f_H^2 E_\alpha - 4 L_Z(T)} T^2$$

- Requires: high temperature ( $T \sim 10\text{-}20$  keV), density, and confinement time
- Magnetic confinement: tokamaks and stellarators



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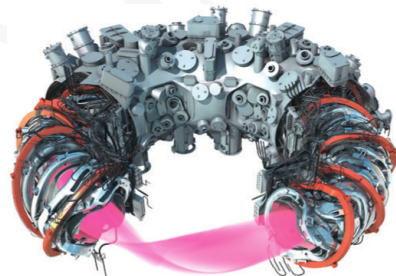
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## Triple Product

Key figure of merit for fusion performance combining density, temperature, and energy confinement time

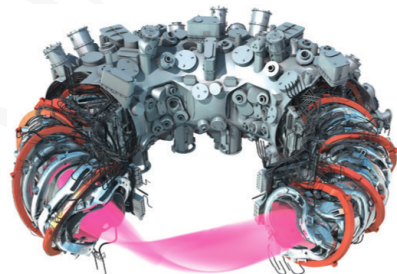
# WENDELSTEIN 7-X STELLARATOR

- World's largest stellarator ( $R = 5.5$  m,  $a \sim 0.5$  m)
- 50 superconducting coils, 5-fold symmetry
- Island divertor configuration
- Optimized for reduced neoclassical transport
- Operational phases: OP1.1, OP1.2a, OP1.2b
- Achieved 30 minute plasma discharges



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## Key Challenge

Managing heat loads on plasma-facing components through controlled radiative cooling

# PLASMA RADIATION AND TRANSPORT

## Transport Mechanisms:

- Classical: collisional diffusion
- Neoclassical: trapped particles,  $\propto 1/\nu$
- Anomalous: turbulent transport (dominant)

## Radiation Processes:

- Bremsstrahlung:  $P_{\text{Brems}} \propto n_e^2 T^{1/2}$
- Line radiation: atomic transitions
- Impurity effects on plasma performance

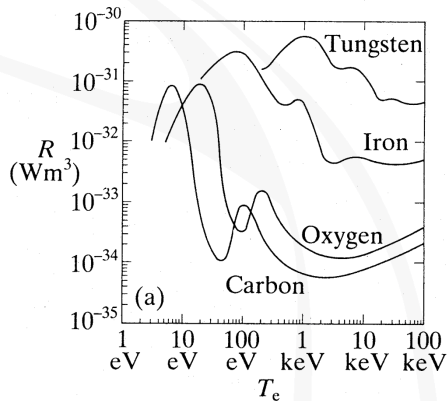
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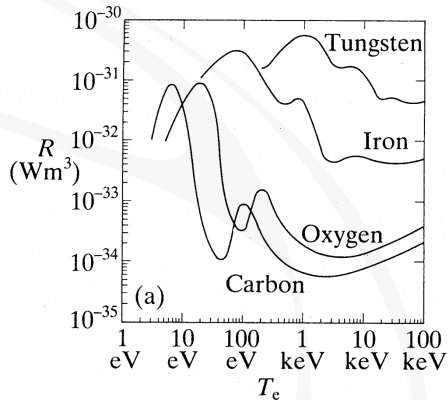
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Deliberate injection of low-Z (He, N<sub>2</sub>) or high-Z (Ne, Ar) impurities to enhance

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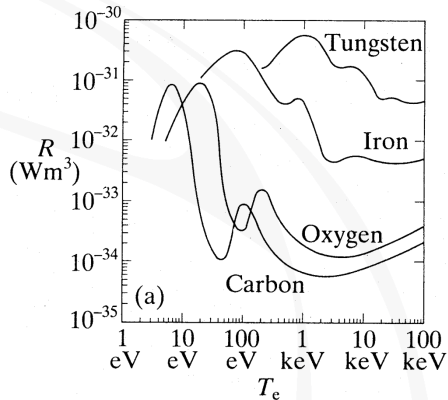
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## Radiation Fraction:

$$f_{\text{rad}} = \frac{P_{\text{rad}}}{P_{\text{tot}}}$$



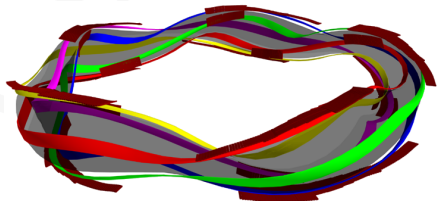
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# ISLAND DIVERTOR CONFIGURATION

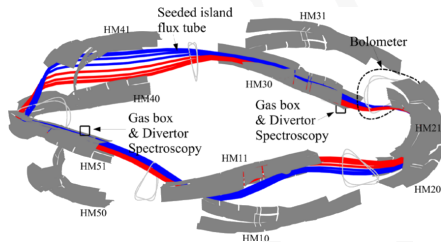
## Magnetic Configuration:

- Standard configuration:  $\iota = 5/5$
- Five helical island chains
- Ten divertor modules (5 half-modules)
- Field lines connect upstream to downstream targets
- Connection lengths:  $\mathcal{O}(100\text{-}1000\text{ m})$



## Gas Injection Strategy:

- Thermal helium beam valves
- Located behind divertor targets
- Direct seeding into island chains
- Fast piezo valves (ms response)



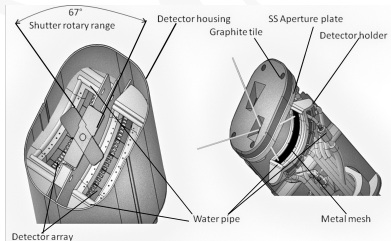


# BOLOMETER DIAGNOSTIC SYSTEM

# BOLOMETRY AT W7-X

## Metal Resistor Bolometers:

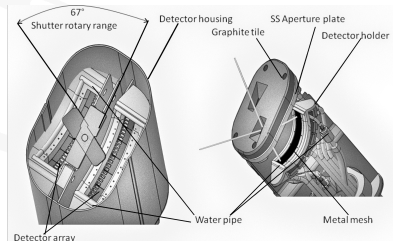
- Thin metal film absorbers (Pt, Au)
- Wheatstone bridge circuit
- Measures total radiation power
- Multicamera system: HBC, VBCI, VBCr
- 128 channels total
- Spatial resolution:  $\sim 5$  cm at magnetic axis



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## Measurement Principle:

$$P_M = F_M \cdot \left( \frac{d(\Delta \tilde{U}_M)}{dt} + f_M \Delta \tilde{U}_M \right)$$

## Global Radiation Power:

$$P_{\text{rad}} = \frac{V_{P,\text{tor}}}{V_C} \sum_M \frac{P_M V_M}{A_M}$$

# BOLOMETER CALIBRATION AND PERFORMANCE

## Calibration Methods:

- In-situ laser calibration
- Electrical calibration
- Etendue calculations:  
$$K_M = \int_M \widetilde{K}_M dA_M$$
- Volume determination:  $V_M$  (LOS cone volume)

## Performance Characteristics:

- Time resolution:  $\sim 1$  ms
- Sensitivity:  $\sim 10$  kW/m<sup>3</sup>
- Uncertainty:  $\sim 10$ -15%

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## Chord Brightness

Line-integrated measurement along each detector's line of sight provides radial profile information

## Applications:

- Power balance calculations
- Radiation profile monitoring
- Real-time feedback control
- Tomographic reconstruction

# REAL-TIME RADIATION FEED- BACK CONTROL

# FEEDBACK SYSTEM DESIGN

## System Architecture:

- NI 6321 data acquisition hardware
- LabVIEW control software
- PID controller implementation
- Thermal gas valve actuation
- Minimum latency: 13.6 ms

## PID Controller:

$$u(t) = K_p e(t) + K_i \int e(t') dt' + K_d \frac{de(t)}{dt}$$



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## Radiation Prediction Proxies:

*Subset-based (3-7 channels):*

$$P_{\text{pred}}^{(1)} = \frac{V_{P,\text{tor}}}{V_S} \sum_M^S \frac{P_M V_M}{K_M}$$

*Single channel (dimensionless):*

$$P_{\text{pred}}^{(2)} \propto \Delta U_M$$

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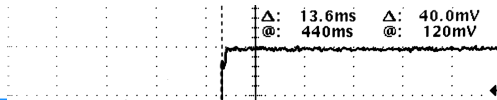
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# EXPERIMENTAL ACHIEVEMENTS

## XP20181010.32 - Benchmark

### Discharge:

- Stable helium-seeded detachment
- Radiation fraction:  $f_{\text{rad}} > 90\%$
- Peak values up to 100%
- Target heat load reduction: factor of 2
- $\text{C}^{3+}$  detachment at  $f_{\text{rad}} \sim 50\%$
- Validated 3-channel LOS subset

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### Key Result

Demonstrated stable, feedback-controlled radiative cooling with  $f_{\text{rad}} \geq 85\%$  without terminal plasma disruption

### Comparison with Other Methods:

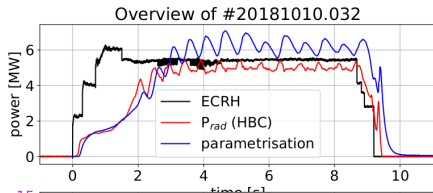
- Electron density feedback
- C-III filterscope feedback
- Bolometer feedback shows intrinsic connection to detachment physics

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# FEEDBACK PERFORMANCE ANALYSIS

- **Gas Injection Dynamics:** Moderately scaled gas puffs (medium length and intensity) most effective for reliable edge cooling
- **System Limitations:** Computational and algorithmic constraints introduce non-negligible latencies
- **Optimization Challenges:** Difficult to optimize during commissioning due to limited experimental time

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## Critical Finding

The intrinsic connection of  $P_{\text{rad}}$  and  $f_{\text{rad}}$  to the detachment process makes bolometer feedback essential for future fusion reactor applications

# LINE OF SIGHT SENSITIVITY AND MODELING



# IMPURITY SEEDING MODELS

## Two-Chamber Model:

$$\dot{N}_w = (N_{w,\text{lim}} - N_w) \tau_{w,p}$$

$$\dot{N}_p = \Gamma_s + N_w \tau_{w,p} - f_{w,p} N_p - N_p \tau_p$$

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- Particle exchange rates
- Pumping and recycling

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## Three-Chamber Model:

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- Adds SOL compartment
- Better represents edge physics
- Both models equally capable of representing feedback measurements

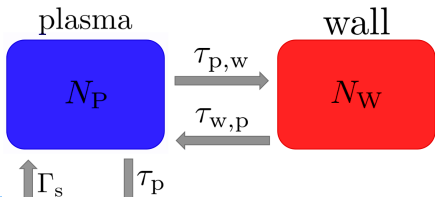
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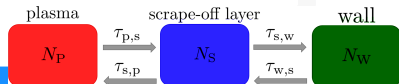
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# CHANNEL SELECTION SENSITIVITY

## LOS Sensitivity Evaluation:

- Weighted deviation metric
- Correlation analysis
- Mean deviation assessment

## Optimal Channel Selection:

- No single "best" set exists
- Robust selection achieves  $\geq 85\%$  prediction accuracy
- Agreement between HBC and VBC cameras

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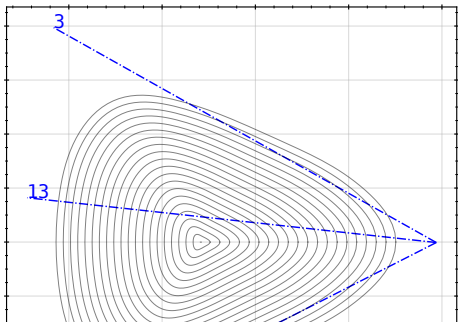
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## Key Finding

Detectors viewing separatrix and SOL region most viable for real-time feedback configurations



# STRAHL IMPURITY TRANSPORT MODELING

## STRAHL Code:

- 1D impurity transport simulation
- Models carbon and oxygen radiation
- Includes ionization, recombination
- Diffusion and convection transport

## Key Results:

- Carbon dominates SOL emissivity
- Reduced diffusivity near separatrix
- Inward shift of radiation for  $f_{\text{rad}} \rightarrow 1$
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## Forward Modeling:

- Calculate chord brightness from STRAHL profiles
- Compare with experimental measurements
- Validate transport assumptions

## Discrepancy

Forward calculations show less asymmetry in chord brightness than experiments, suggesting model limitations in capturing 3D effects

# TOMOGRAPHIC RECONSTRUCTION



# MINIMUM FISHER REGULARIZATION (MFR)

## Ill-Posed Inverse Problem:

- 128 line-integrated measurements
- Reconstruct 2D emissivity:  $\sim 4500$  pixels
- Requires regularization

## Fisher Information:

$$I_F = \int \frac{1}{g(\vec{r})} \left( \frac{\partial g(\vec{r})}{\partial \vec{r}} \right)^2 d\vec{r}$$

## Iterative Solution:

$$\vec{x}^{(n+1)} = \left( \mathbf{T}^T \mathbf{T} + \mu \mathbf{H}^{(n)} \right)^{-1} \mathbf{T}^T \vec{b}$$

## Radially Dependent Anisotropy (RDA):

- Tailored weighting:  $k_{\text{ani}}(r)$
- $k < 1$ : localized structures
- $k > 1$ : smooth distributions
- Separate core and edge parameters

## MFR Advantages

- Robust to noisy data
- Smooth solutions
- Incorporates a priori knowledge
- Stable convergence

# GEOMETRY SENSITIVITY AND BENCHMARKING

## Geometry Perturbation Tests:

- Camera position variations
- Detector/aperture segmentation  
( $N=2,4,8$ )
- Triangulation vs rectangular splitting
- Etendue calculation validation

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## Key Findings:

- Significant robustness to geometry variations
- $N \geq 8$  segmentation adequate
- Intrinsic bias toward upper SOL/separatrix
- Camera displacement effects non-negligible

# GEOMETRY SENSITIVITY AND BENCHMARKING

## Phantom Image Benchmarks:

- Simple and complex test profiles
- Quality metrics:  $\chi^2$ ,  $\rho_c$
- Optimal  $k_{\text{ani}}$  depends on profile
- No universal "best" parameters

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## Recommended Settings

$k_{\text{ani}} = \{2.0, 0.3\}$  (core, edge)

Grid:  $30 \times 150$  pixels

Domain:  $1.3^2 V_P$

# EXPERIMENTAL TOMOGRAPHY RESULTS

## Statistical Analysis:

- Forward vs backward  $P_{\text{rad}}$  correlation
- Core radiation slightly overestimated
- Total power significantly overestimated
- Consistent with phantom benchmarks

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## Power Balance Application:

$$P_{\text{bal}} = P_{\text{ECRH}} - P_{\text{rad}} - P_{\text{div}}$$

- $P_{2\text{D}}^{\text{core}}$  improves balance accuracy
- More stable than  $P_{\text{rad}}$  alone
- Alternative for quasi-steady-state analysis

# TOMOGRAPHY QUALITY ASSESSMENT

- **Reconstruction Accuracy:**  $P_{\text{rad}}$  consistently underestimates total 2D integrated power by amount equal to SOL radiation
- **Quality Metrics:**  $\chi^2$  and correlation coefficient  $\rho_c$  not always congruent - optimal tomogram may not have  $\chi^2 = 1$
- **Anisotropy Optimization:** Ideal  $k_{\text{ani}}$  set exists for each profile but dimensionality makes finding it impractical
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## Conclusion

MFR with RDA weighting provides robust, reliable 2D radiation reconstructions suitable for physics analysis and power balance calculations, though computational limitations prevent continuous real-time operation

# CONCLUSIONS AND OUTLOOK

# KEY ACHIEVEMENTS

1. **Real-Time Feedback System:** Successfully implemented, tested and operated first bolometer radiation feedback at W7-X
2. **Line of Sight Optimization:** Systematic sensitivity analysis identified optimal detector configurations
3. **Tomographic Reconstruction:** Developed and benchmarked tailored MFR algorithm with RDA weighting

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3. **Tomographic Reconstruction:** Developed and benchmarked tailored MFR algorithm with RDA weighting
  - Robust to geometry perturbations and noise
  - Successfully applied to experimental data
  - Provides alternative for power balance calculations

# SCIENTIFIC INSIGHTS

## Detachment Physics:

- $C^{3+}$  detachment visible at  $f_{\text{rad}} \sim 50\%$
- Stable detachment achievable at  $f_{\text{rad}} > 90\%$
- Radiation shifts inward as  $f_{\text{rad}} \rightarrow 1$
- X-point and island radiation concentration

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## Feedback Control:

- Moderate gas puffs most effective
- No simple scaling law found
- Intrinsic connection to detachment essential
- System latency limits performance

# SCIENTIFIC INSIGHTS

## Detachment Physics:

- $C^{3+}$  detachment visible at  $f_{\text{rad}} \sim 50\%$
- Stable detachment achievable at  $f_{\text{rad}} > 90\%$
- Radiation shifts inward as  $f_{\text{rad}} \rightarrow 1$
- X-point and island radiation concentration

## Impurity Transport:

- Carbon dominates SOL emissivity
- Reduced diffusivity near separatrix
- Complex 3D effects in stellarator geometry

## Feedback Control:

- Moderate gas puffs most effective
- No simple scaling law found
- Intrinsic connection to detachment essential
- System latency limits performance

## Diagnostic Performance:

- Multicamera system provides comprehensive coverage
- Geometry uncertainties affect results
- Tomography reveals SOL radiation underestimation

# FUTURE OUTLOOK AND RECOMMENDATIONS

## System Upgrades:

- Reduce feedback latency through hardware/software optimization
- Implement more complex  $P_{\text{pred}}$  models
- Per-experiment configurability
- Develop predictive scaling laws

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- Integrate with other SOL models
- Better parametric optimization
- 3D effects in STRAHL modeling

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- Larger phantom benchmark set
- Additional virtual camera arrays
- Real-time capable algorithms
- Automated parameter optimization

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## Reactor Relevance:

- Essential for DEMO power plant operation
- Reliable radiative cooling at  $f_{\text{rad}} \geq 95\%$
- Machine protection and heat load management
- Steady state operation capability

# BROADER IMPACT

## Contribution to Fusion Energy Research

This thesis advances the state-of-the-art in plasma diagnostics and control for stellarator fusion devices, providing essential tools and knowledge for achieving reactor-relevant operating scenarios with controlled radiative cooling and stable detachment.

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- First real-time bolometer feedback at W7-X
- Validated diagnostic optimization methods
- Benchmarked tomography algorithm
- Demonstrated stable high- $f_{\text{rad}}$  operation



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- Benchmarked tomography algorithm
- Demonstrated stable high- $f_{\text{rad}}$  operation

### Scientific Contributions:

- Detachment physics understanding
- Impurity transport insights
- Radiation distribution characterization
- Power balance methodology

# SUMMARY

- Developed and operated first real-time bolometer radiation feedback system at W7-X stellarator
- Achieved stable, controlled detachment with radiation fractions  $f_{\text{rad}} \geq 85\%$  and peaks up to 100%
- Reduced divertor target heat loads by factor of 2 through controlled impurity seeding
- Validated 3-channel bolometer subset for fast radiation power prediction with  $\geq 85\%$  accuracy
- Systematic line of sight sensitivity analysis identified optimal detector configurations for feedback

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- Systematic line of sight sensitivity analysis identified optimal detector configurations for feedback
- STRAHL modeling revealed carbon dominance in SOL radiation and inward shift at high  $f_{\text{rad}}$
- Developed and benchmarked Minimum Fisher Regularization tomography with radially dependent anisotropy
- Demonstrated robust 2D radiation reconstruction from experimental data
- Provided alternative power balance methodology using tomographic integrals

# COLLABORATIONS

## Acknowledgement

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## Collaborators:



**HELMHOLTZ**  
RESEARCH FOR GRAND CHALLENGES



**EUROfusion**

Thank You

# REFERENCES I

# Backup Slides

Additional Details

# MEASUREMENT ALGORITHM OVERVIEW

## Algorithm Stages:

### 1. Initialization (T1-60s)

- Load geometry matrices  $K_M, V_M$
- Configure channel selection  $S$
- Initialize analog output

### 2. Calibration (4s)

- Offset measurement
- Two-stage calibration
- Determine  $R, \tau, \kappa$

### 3. Measurement

- Continuous voltage integration
- Moving average filtering
- Real-time prediction

## Real-time Feedback:

- Parallel analog output via NI 6321
- Two prediction channels  $P_{\text{pred}}^{(1)}, P_{\text{pred}}^{(2)}$
- FIFO buffer management
- Sample time  $\Delta t$  configurable

## Data Storage:

- Local HDD backup
- W7-X archive upload
- Calibration parameters
- Full voltage traces



# FOURIER TRANSFORM CORRELATION METRIC

## Alternative LOS Sensitivity Evaluation:

### Method:

- Uses spectral correlation between  $P_{\text{rad}}$  and  $P_{\text{pred}}$
- Discrete Fast Fourier Transform (DFFT)
- Cross-spectral density analysis
- Coherence-based metric

### Mathematical Formulation:

$$\gamma_i = \mathcal{F}(P_{\text{rad}})(\omega)$$

$$\gamma_j = \mathcal{F}(P_{\text{pred}})(\omega)$$

$$\varphi(\omega) = \sqrt{\frac{|g_{i,j}^2|}{|\gamma_i \gamma_j|}}$$

### Key Results:

- Similar patterns to correlation metric
- Global maximum at upper inboard X-point
- HBC: best at  $0.38r_a$
- VBC: best at  $-0.13r_a$
- Spectral focus excludes absolute value contributions

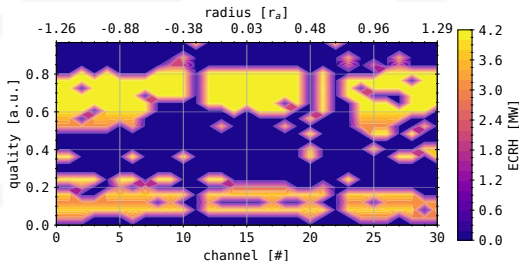
### Advantages:

- Frequency domain analysis

Robust to noise

# PLASMA PARAMETER SENSITIVITY ANALYSIS

## Correlation with Plasma Parameters:



Heating power  $P_{\text{ECRH}}$  correlation

## Key Findings:

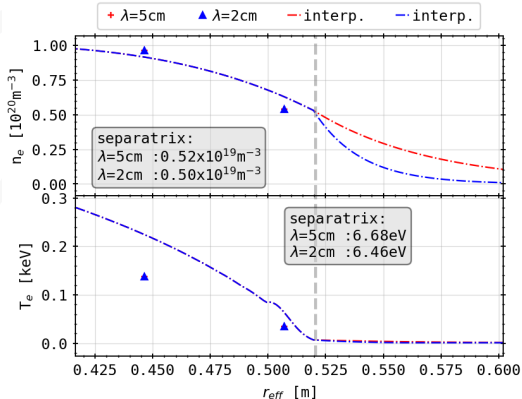
- Higher  $P_{\text{ECRH}}$  ( $\geq 3.5$  MW)  $\rightarrow$  better predictability
- Optimal  $f_{\text{rad}} > 0.75$
- Core  $T_e \sim 1.3$  keV ideal
- Gap at  $\vartheta \approx 0.3-0.5$
- Both cameras in good agreement

## Optimal Discharge Conditions:

- Medium-high heating power
- High radiation fraction
- Moderate core temperature
- Consistent across LOS combinations

# STRAHL PARAMETER VARIATIONS

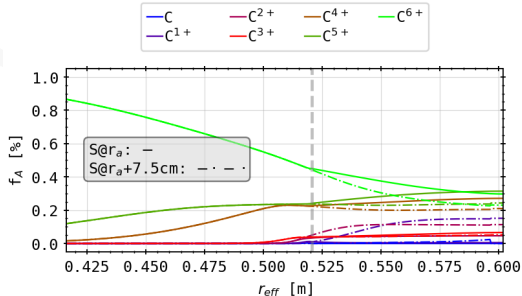
## Decay Length Variation:



SOL decay length:  $\lambda = 2\text{cm}$  vs  $5\text{cm}$

## Impact:

## Impurity Source Location:

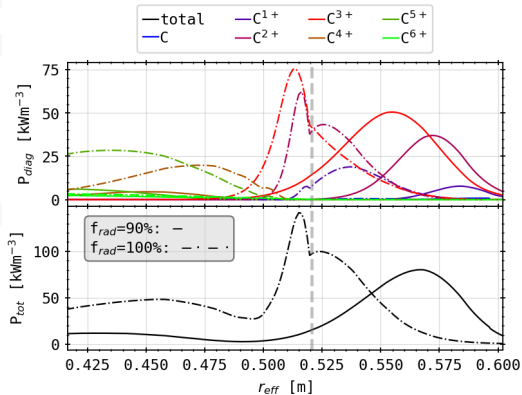


Source at LCFS vs divertor ( $r_a+7.5\text{cm}$ )

## Impact:

- Changes ion populations in SOL
- Similar core emissivity results
- Validates model robustness

# STRAHL: RADIATION FRACTION DEPENDENCE



Total radiation for  $f_{\text{rad}} = 90\%$  and  $100\%$

## Critical Observations:

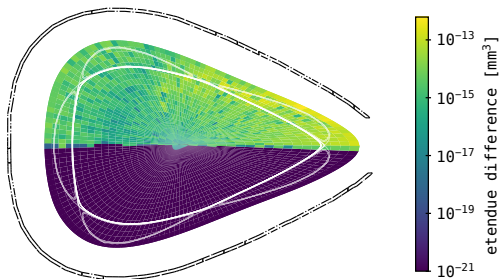
- At  $f_{\text{rad}} = 90\%$ : peak in SOL
- At  $f_{\text{rad}} = 100\%$ : peak shifts to LCFS
- $\text{C}^{3+}$  and  $\text{C}^{2+}$  most sensitive
- Dramatic profile changes near separatrix

## Implications:

- LOS sensitivity strongly depends on  $f_{\text{rad}}$
- Detachment scenarios predictable
- Validates feedback system design
- Critical for controlled detachment

# CAMERA GEOMETRY SENSITIVITY

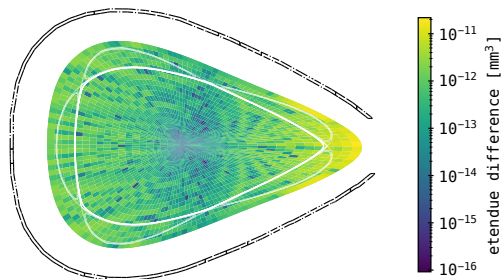
## Unilateral Mirroring:



Etendue change:  $<10^{-13}$  permm

- Half of HBC array mirrored
- Minimal impact on forward model
- Asymmetry reduced by 0.5-1%

## Artificial Symmetric Camera:



Fully symmetric, equidistant array

- Perfectly symmetric geometry
- Similar etendue variations
- Validates as built design

# TOMOGRAPHY: RDA VS RGS COMPARISON

## Radially Dependent Anisotropy (RDA):

- Standard MFR with  $k_{\text{ani}}$  profile
- Smooth reconstructions
- Good radial profile matching
- Established method

## Relative Gradient Smoothing (RGS):

- Weight:  $W_{\text{RGS}} = (1/g_i)^2$
- Can be combined with RDA
- Stronger feature localization
- Higher local intensities

## Comparison Results:

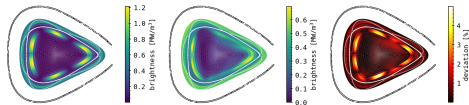
- Similar integrated power values
- RGS: sharper features
- RGS: better peak localization
- RGS: may lose weak structures
- RDA: more robust for general use

## Conclusion:

- RDA sufficient for most cases
- RGS useful for specific scenarios
- Combined approach possible
- Choice depends on physics goals

# PHANTOM TOMOGRAPHY: EDGE CASES

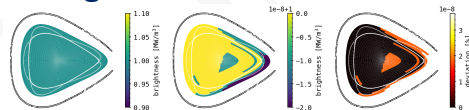
## Inverted Anisotropy:



Bright SOL ring + core islands

- More challenging reconstruction
- SOL dominates measurement
- Core features partially obscured

## Homogeneous Distribution:



Uniform 1 MW/m<sup>3</sup> everywhere

- Tests baseline reconstruction
- Validates algorithm behavior
- Confirms no artificial structures

**Lessons:** Camera sensitivity in SOL can obscure core features; algorithm performs well for uniform distributions; inverted scenarios more challenging than standard core-peaked profiles.

# APPENDIX SUMMARY

## Measurement Algorithm:

- Three-stage process validated
- Real-time capability confirmed
- Robust calibration procedure
- Full data archiving

## LOS Sensitivity Methods:

- Fourier correlation complements time-domain
- Plasma parameter correlations identified
- Optimal discharge conditions determined

## STRAHL Validation:

- Parameter variations tested
- Model robustness confirmed
- Critical  $f_{\text{rad}}$  dependence shown

## Tomography Extensions:

- Geometry perturbations minimal
- RDA vs RGS compared
- Edge case phantoms tested
- Algorithm limits characterized